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A Review of Magnetic Stratigraphy Investigations in Cretaceous Pelagic Carbonate Rocks

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Pelagic carbonate rocks possess many suitable characteristics for paleomagnetic and magnetostratigraphic studies. Paleomagnetic results are summarized for seven lengthy sections of pelagic limestones and marls from Umbria and the southern Alps in Italy. Differences in apparent polar wander paths from these two regions are interpreted in terms of tectonic rotation of allochthonous Umbria. The magnetic stratigraphies of the paleontologically dated sections are independent of their tectonic differences and are combined to form a continuous record of geomagnetic polarity for the Barremian through Maastrichtian stages of the Cretaceous. All but one of the reversals in these sections are confirmed by duplication in at least one other section. Additional Cretaceous reversals have been reported in other land sections and in DSDP (Deep Sea Drilling Project) and IPOD (International Program of Ocean Drilling) cores. Some of these reversals are not defined well magnetically, and confirmation of others is clouded by imprecise paleontological dates. If real, they are probably of short duration. The confirmed reversal sequence correlates well with the Cretaceous oceanic magnetic anomaly sequence. The ages of certain key anomalies are altered: Late Cretaceous anomalies 29–34 are younger, and Early Cretaceous anomalies *M*0 and *M*1 are older than previously thought. The longer duration of the Cretaceous magnetic quiet interval of normal polarity results in a reduction of corresponding sea floor spreading rates to about 70% of earlier values, but they are still appreciably higher than during formation of the preceding *M* sequence anomalies.

INTRODUCTION

The geomagnetic polarity sequence in the Cretaceous has been determined from two sources: from paleomagnetic studies on paleontologically or radiometrically dated rocks and from analysis of lineated oceanic magnetic anomalies.

On the basis of a summary of all available paleomagnetic data from continental rocks, *Helsley and Steiner* [1969] deduced the existence of a long period of constant normal geomagnetic polarity during the Cretaceous from 85 to 110 m.y. The relative dating of isolated Cretaceous sampling sites is too imprecise, because of the limited resolution of radiometric and paleontological dating methods, to allow development of a magnetic stratigraphy for the mixed polarity intervals preceding and following this period. As a sequence of normal and reversed polarities is essentially a binary signal, correlation of land sections with each other and with the oceanic anomalies depends on the recognition of patterns of polarity reversals. Stratigraphically continuous sedimentary sequences covering long periods of time are desirable for the recognition of correlatable patterns.

The marine magnetic record has provided our best information about the history of the geomagnetic field during the Cretaceous (and also most of the Cenozoic). The pattern of oceanic magnetic lineations defines three distinct episodes of differing character of the geomagnetic field during the Cretaceous. The Late Cretaceous interval of mixed polarities corresponds to the oldest of the numbered sequence of anomalies that stretches through the Cenozoic to the present time [*Heirtzler et al.*, 1968]. The Early Cretaceous is characterized by the youngest part of the Mesozoic anomaly sequence [*Larson and Pitman*, 1972]. The two magnetic anomaly sequences are separated in the oceans by the Cretaceous magnetic quiet zone, within which are found anomalies whose amplitudes are generally subdued, and which cannot be correlated to form lineations. They correspond to the interval of constant geomagnetic polarity inferred from the continental paleomagnetic record.

In this paper a number of detailed land-based and DSDP (Deep Sea Drilling Project) paleomagnetic studies that have been published elsewhere are integrated and evaluated with the objective of summarizing the present state of knowledge of geomagnetic polarity during the Cretaceous. Magnetic stratigraphy in several sequences of pelagic carbonate rocks in Umbria and the southern Alps of northern Italy are used to correlate and paleontologically date much of the Cretaceous magnetic reversal sequence interpreted from oceanic magnetic anomalies.

SEDIMENTOLOGICAL AND MAGNETIC PROPERTIES OF PELAGIC CARBONATE ROCKS

The pelagic carbonate rocks of northern Italy possess many suitable characteristics for magnetic stratigraphy studies. They were deposited in deep water on the southern margin of Tethys in generally tranquil conditions that persisted through most of the Cretaceous and early Tertiary. Today, stratigraphic thicknesses of several hundred meters of continuously deposited limestones and marls are available for paleomagnetic sampling.

The carbonate matrix of micritic coccolith material accumulated slowly and steadily and contained a varying influx of wind-borne and water-transported detrital particles. These consisted largely of clay minerals and a miniscule fraction of ferrimagnetic minerals which acquired the paleomagnetic field direction by a mechanism of postdepositional alignment [*Irving and Major*, 1964; *Kent*, 1973].

Bioturbation is a characteristic feature of the limestones,

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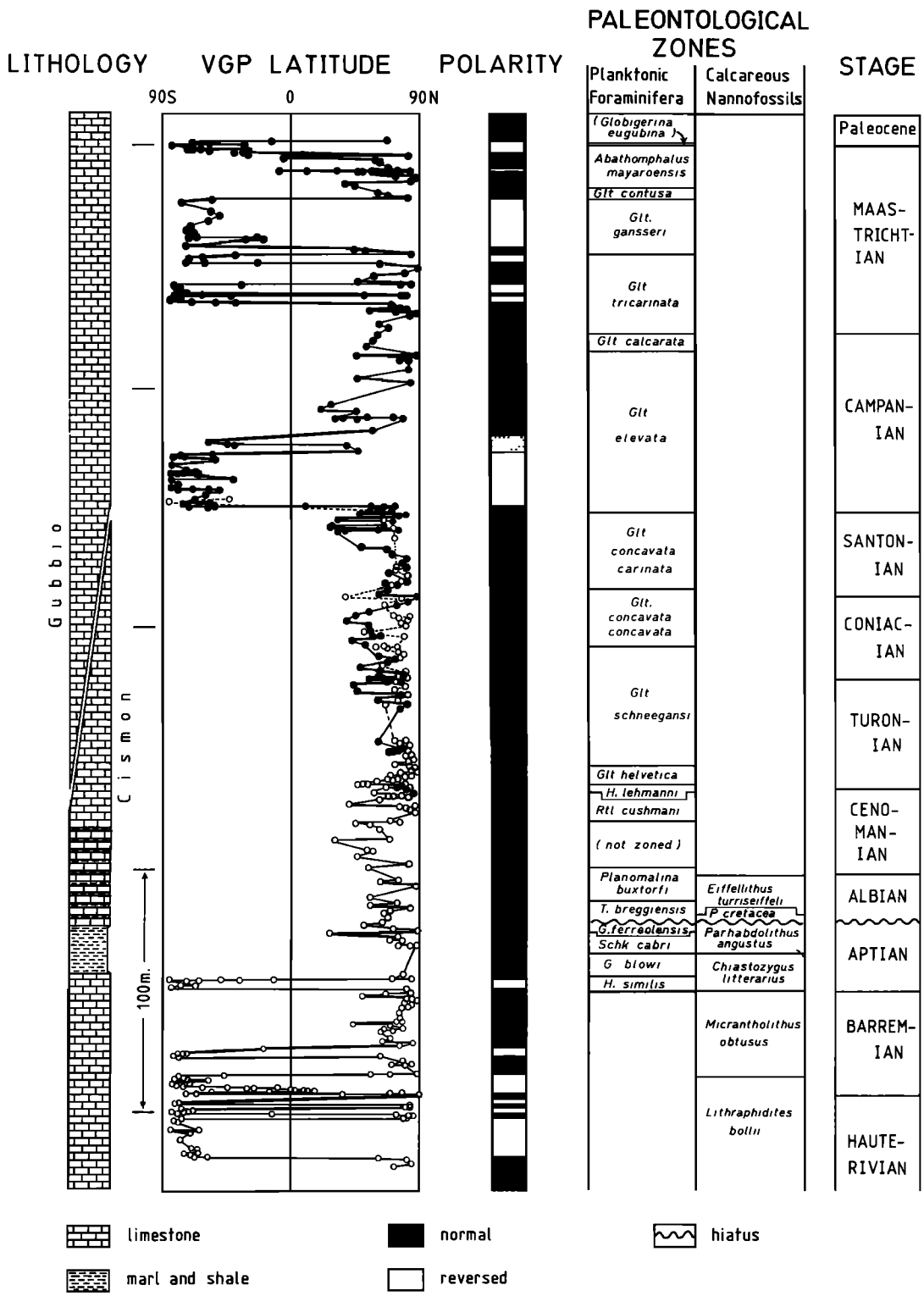


Fig. 1. A summary of combined magnetostratigraphic and biostratigraphic investigations for the pelagic limestone sections at Gubbio (Umbrian Apennines) and Cismon (southern Alps) in Italy. The relative VGP latitudes (see text) for Gubbio (solid circles) and Cismon (open circles) have been superposed to give a composite polarity column. (A stippled polarity zone in the Campanian of the Gubbio section corresponds to tectonically deformed rocks with unreliable magnetization directions.)

and depositional features on a scale finer than about 10 cm have been blurred or destroyed. After bioturbation the sediment is still soft enough for postdepositional magnetization to occur [*Kent*, 1973]. The magnetization is locked into the sediment a few centimeters below the water-sediment interface or base of the bioturbated zone, and a time lag ensues between depositional age and magnetization age [*Løvlie*, 1974]. This is probably of the order of 10,000 years in the rocks we have studied. Short reversals are preserved, because the magnetization is acquired after bioturbation.

The pelagic carbonate sequences can be dated quite precisely using the planktonic foraminiferal zonation schemes for the Aptian to Maastrichtian stratigraphic stages (Figure 1). Calcareous nannofossils must be used for the Barremian and

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TABLE 1. Average Sedimentation Rates Within Each Paleontological Stage in Some of the Umbrian Pelagic Carbonate Sections

Magnetostratigraphic Section	Geological Formation	Paleontological Stage	Age, m.y.	Thickness, m	Sedimentation Rate, m/m.y.
Gubbio, Moria	Scaglia Rossa limestone	Maastrichtian	65	77	15.4
			70		
		Campanian		74	9.2
			78		
		Santonian		36	9.0
			82		
		Coniacian		24	6.0
			86		
		Turonian		34	5.7
			92		
Moria	Scaglia Bianca	Cenomanian		77	9.6
			100		
Poggio le Guaine	Fucoid marls	Albian		72	ca.6
			108		
		Aptian			
Gorgo a Cerbara	Majolica limestone	Barremian	115	57	ca.6
			121		

Ages of the stage boundaries are from *Van Hinte* [1976].

Hauterivian stages, and calpionellids for older Cretaceous stages. Hiatuses are hard to identify unless they are sufficiently prolonged to be identifiable from the biostratigraphic record. In the Cismon section [*Channell et al.*, 1979] from the southern Alps the Early Albian foraminiferal zone of *Ticinella bejaouaensis* is completely missing (Figure 1).

The sedimentation rates are rather inexactly determined because the 'absolute' dating by association of numeric ages with stage boundaries is still imprecise. Typically, however, the rates range around 6–15 m/m.y. (Table 1). Within each standard 1-inch paleomagnetic sample the paleomagnetic field directions are averaged for about 1500–5000 yr; most secular variation effects ought to be averaged out.

A fairly large amount of directional scatter is often observed. In part this may be due to a rather inefficient magnetization process. But also it represents the technical difficulty of accurately measuring and identifying the stable component in each of the hundreds of weakly, partially unstably magnetized specimens that are needed to give a detailed magnetic stratigraphy. The weakness of the remanence is the primary reason for the neglect of pelagic carbonate rocks in paleomagnetic studies until recently. The average intensity of the natural remanence in the Scaglia Rossa limestone is 2×10^{-3} A/m; in the Majolica (Biancone) limestone it is 5×10^{-5} A/m.

The characteristic remanence (ChRM) in these pelagic carbonate rocks is carried predominantly by magnetite, with a concentration around 0.01% in the Scaglia Rossa [*Lowrie and Alvarez*, 1975] and correspondingly lower in the Majolica. Hematite is also a common constituent. It gives the red color to certain 'couches rouges' beds in the Umbrian Fucoid marls, and the pink hue to the Scaglia Rossa. It apparently formed in the Umbrian Scaglia Rossa penecontemporaneously with deposition and burial, but in Scaglia Rossa samples from Sicily it formed much later [*Channell*, 1978]. The diagenetic process responsible is only loosely understood. Rock magnetic studies to determine the significance of possible secondary magnetizations are therefore an inseparable accompaniment to paleomagnetic investigations in pelagic carbonates.

PALEOMAGNETIC RESULTS FROM SECTIONS IN UMBRIA AND THE SOUTHERN ALPS

The moderately weakly magnetized Scaglia limestones were readily measureable with most modern magnetometers; for proper determination of magnetization direction in weakly magnetized Majolica (Biancone) samples a cryogenic magnetometer [*Goree and Fuller*, 1976] was essential. In each section investigated the ChRM was determined after progressive alternating field (AF) or thermal demagnetization. ChRM directions were determined with the aid of several techniques; vector diagrams, a directional stability index, or the minimum dispersion criterion.

Apparent Polar Wander: Tectonic Implications

At each sampling locality a simple correction for local bedding attitude was made by rotating the ChRM directions about a horizontal axis (the strike line). This overlooks the effects of variability of direction and plunge of fold axes, which can strongly influence paleomagnetic directions and the paleotectonic implications inferred from them. However, the plunge of folds is seldom more than a few degrees, so bedding corrections about the strike line probably do not introduce significant errors.

An apparent polar wander (APW) path from a single section will be more strongly affected by local tectonics than an APW path derived by combining data from several sections. *VandenBerg et al.* [1978] developed an Umbrian APW path from Late Jurassic to Eocene by combining data from a few localities according to paleontological age. After correcting for a late Tertiary tectonic rotation they found excellent agreement between their Umbrian APW path and the 'African' APW path of *Van der Voo and French* [1974].

Irving [1977] has derived smoothed APW paths for North America, Eurasia, and Gondwana since the Devonian by computing moving averages of the best available paleomagnetic data within a 30–40 m.y. window. The Gondwana APW path is necessarily pre-Cretaceous. Following *VandenBerg* [1979] a 'pseudo African' APW path has been constructed by rotating the North American poles about the appropriate rotation poles for the opening of the north Atlantic [*Sclater et al.*, 1977]. Relevant segments of this 'African' APW path and Irving's Eurasian APW path are shown in Figure 2.

We have combined the paleomagnetic data from our Umbrian and southern Alps sections for each Cretaceous stage from Barremian to Santonian (Table 2). The Umbrian directions have distinctly westerly declinations compared to those from the southern Alps. The Cretaceous APW path segments obtained from these directions display larger amplitude wandering than Irving's smoothed paths (Figure 2). Within each region the Barremian data are particularly divergent.

It is clear that the data from the southern Alps have an African affinity [*Channell and Tarling*, 1975], whereas the Umbrian data do not agree with either Eurasian or African data [*Lowrie and Alvarez*, 1975]. Differing interpretations of this discrepancy have been proposed. *VandenBerg* [1979] assumes autochtony of the Umbrian region and explains the discrepancy as late Tertiary counterclockwise rotation of the Italian peninsula about a pole near Malta.

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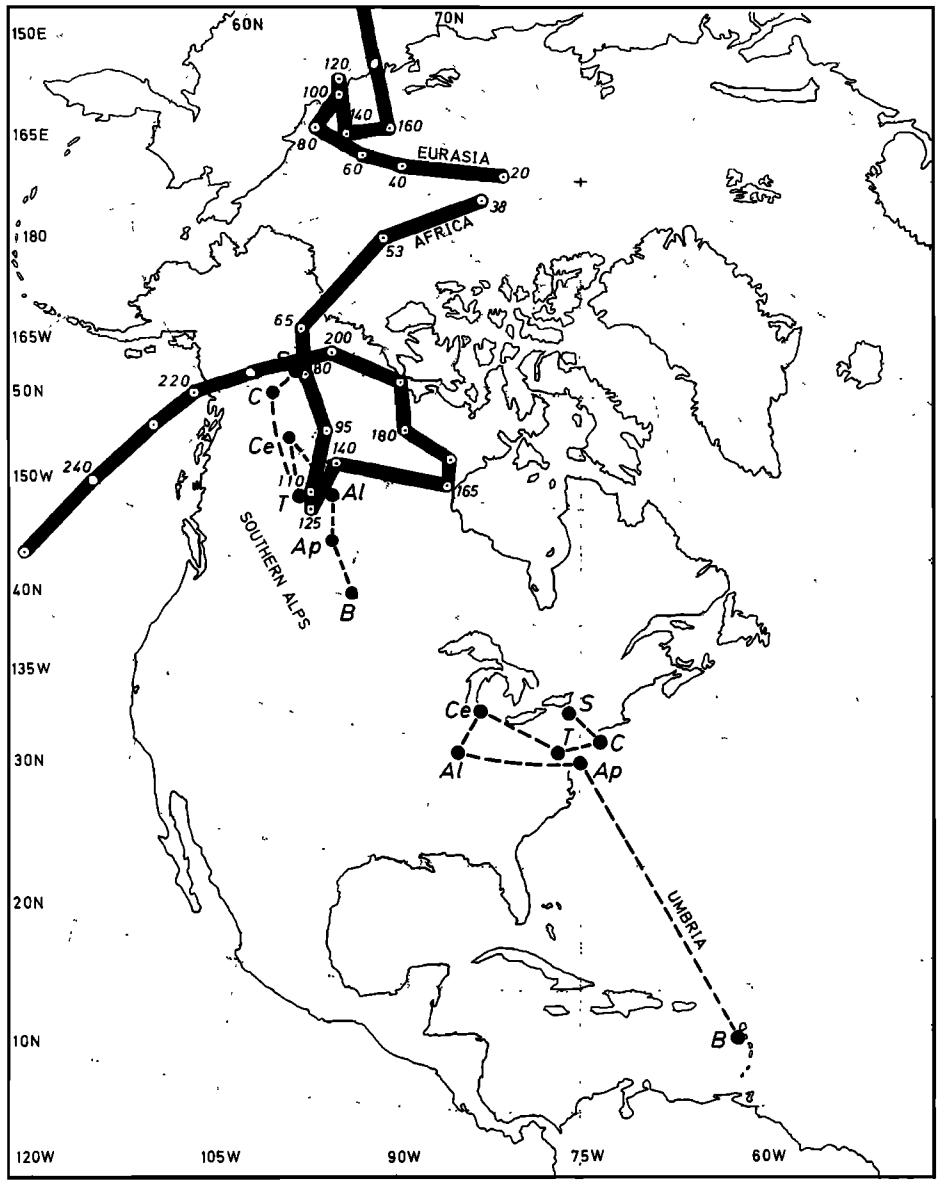


Fig. 2. Barremian-to-Santonian apparent polar wander (APW) paths for magnetostratigraphic sections in Umbria and the southern Alps lie far from the Eurasian APW path of *Irving* [1977]. The Barremian poles are unusual in both regions. Otherwise, the southern Alpine data lie close to the 'pseduo African' APW path (see text), while the Umbrian data do not. (Explanation of symbols: S = Santonian, C = Coniacian, T = Turonian, Ce = Cenomanian, Al = Albian, Ap = Aptian, B = Barremian.)

However, paleomagnetic directions from northern and southern Umbria do not agree with each other but follow fold axis trends [*Channell et al.*, 1978]. Umbria is a region of folding and thrust faulting [*Alvarez et al.*, 1978] with probable oroclinal bending of the stratigraphic section above an Upper Triassic evaporite unit [*Channell et al.*, 1978]. The displacement of the Umbrian APW path in Figure 2 is most likely an expression of the allochthony of the northern Umbrian region where the sections are located.

Determination of Paleomagnetic Polarity in Tectonically Rotated Regions

An objective guide to paleomagnetic polarity was obtained for each section by combining the optimum declination and inclination of ChRM to calculate a relative VGP latitude for each sample. In this computation the latitude of the site during deposition was derived from the inclination of the mean direction in the section, and the declinations were corrected for subsequent tectonic rotations and/or polar wander by referring them to the declination of the site mean direction. The effects of tectonic rotation on magnetic stratigraphy are minimal when this is based on relative VGP latitude, and data from sections with quite different tectonic styles can be conveniently combined.

Figure 1 shows a superposition of relative VGP latitudes from the Gubbio section [*Lowrie and Alvarez*, 1977*b*] in the folded, possibly allochthonous northern Apennines, and from the Cismon section [*Channell et al.*, 1979] in the block-faulted, autochthonous southern Alps. The magnetic stratigraphies show good agreement where they overlap and otherwise complement each other. They yield a sharply defined, unambiguous magnetization reversal sequence.

CRETACEOUS MAGNETIC STRATIGRAPHY

The magnetic stratigraphy has been evaluated in seven partially overlapping sequences of carbonate rocks, five in Umbria and two in the southern Alps (Figure 3). Each time inter-

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TABLE 2. Paleomagnetic Directions Averaged by Paleontological Stage for all the Sections in Umbria and the Southern Alps

Paleontological Stage	Umbria					Southern Alps				
		Mean Direction					Mean Direction			
	No.	Decl.	Incl.	α_{95}	k	No.	Decl.	Incl.	α_{95}	k
Santonian	93	302.0	46.2	4.7	10.6	9	343.0	41.1	10.5	25.1
Coniacian	40	299.0	46.5	6.5	13.1	22	342.5	38.8	4.8	43.0
Turonian	64	301.8	43.8	5.8	10.3	48	331.4	37.2	4.5	22.3
Cenomanian	51	307.7	38.2	3.9	27.7	25	337.3	38.4	10.5	8.5
Albian	25	305.5	35.0	5.7	27.0	50	331.1	41.4	5.5	14.2
Aptian	177	302.8	44.4	3.6	9.6	70	327.8	37.9	5.1	12.1
Barremian	45	272.6	40.1	8.3	7.5	82	321.7	38.1	7.0	6.0
Overall averages										
By samples	495	301.1	43.6	2.1	10.4	306	329.8	38.8	2.6	10.3
By stages	7	300.1	42.6	6.0	101.4	7	333.5	39.2	4.7	166.3

No. = number of samples (stages); Decl., Incl. = declination and inclination, respectively, of the mean direction; α_{95} = radius of 95% confidence circle; k = precision parameter.

val from the Barremian to the Cretaceous/Tertiary boundary is covered in at least two sections. The magnetic stratigraphy of the Cismon section extends through the Hauterivian into the top of the Valanginian [*Channell et al.*, 1979], but only the portion which overlaps the other sections is shown here. Within the negative polarity zone at the base of the Barremian are found three short normal polarity zones, which do not occur in this interval in the overlapping Gorgo a Cerbara section. A hiatus in the Cismon and Cerro Veronese sections is covered in the Valdorbia and Poggio le Guaine sections. A short negative polarity zone in the Valdorbia section is not found in overlapping segments of the other sections. The other reversals in Figure 3 are confirmed by duplication in other sections, thus fulfilling a fundamental credibility criterion. A composite magnetic stratigraphy for the middle and Upper Cretaceous has been produced by compiling the confirmed reversals (Figure 4).

Reversed Directions in other Paleomagnetic Sections

The reversal in the Valdorbia section was first reported by *VandenBerg et al.* [1978], whose site designation ISEA is used in Figure 4. The ISEA direction was incorporated in a composite polar wander path for Umbria. The ISEA reversal occurs in a 'couches rouges' section of the Fucoid marls. Alternating field and thermal demagnetization [*VandenBerg et al.*, 1978; *Lowrie et al.*, 1980] isolated a stable reversed magnetization component. This reversal is possibly not contemporaneous with deposition; it is not found in overlapping sections of the same lithology (Figure 3).

VandenBerg and Wonders [1980] reported two reversals in the Upper Albian in the Valle del Mis (VM) section in the southern Alps. In the Quero-Schievenin section (QS) they found three short reversed polarity zones within a long inter-

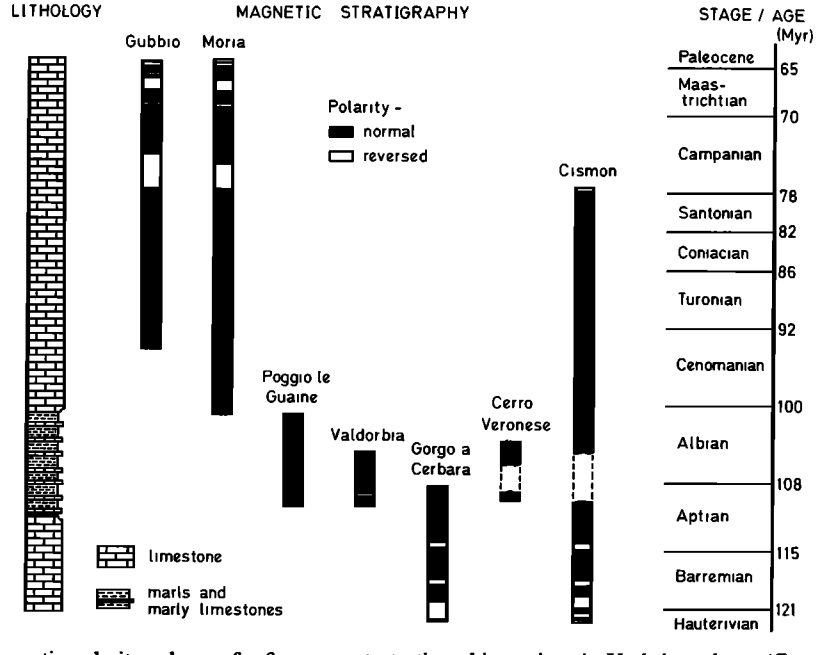


Fig. 3. Geomagnetic polarity columns for five magnetostratigraphic sections in Umbria and two (Cerro Veronese and Cismon) in the southern Alps cover all the Cretaceous stages since (and including) the Barremian. Except for a Late Aptian reversal in the Valdorbia section, the polarity sequence in each section is confirmed in at least one other overlapping section. Numeric ages of stage boundaries are from *Van Hinte* [1976].

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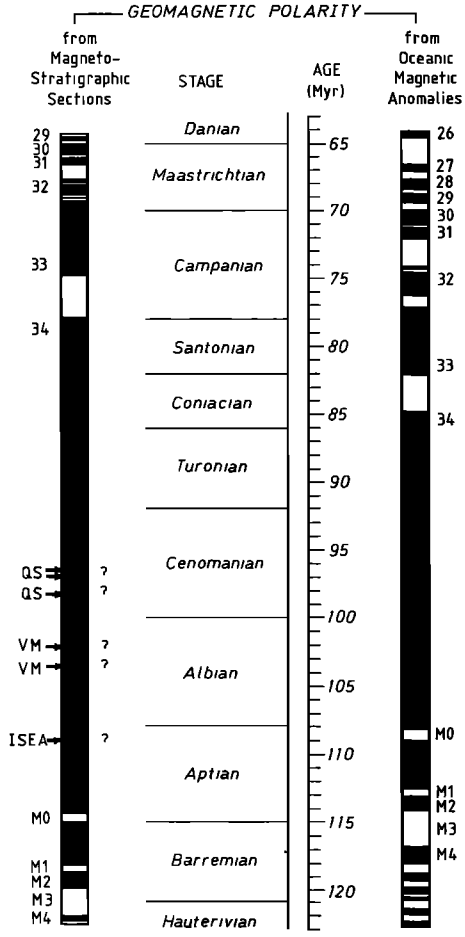


Fig. 4. Comparison of the geomagnetic polarity sequences in the Cretaceous as derived from magnetostratigraphic sections (Figure 3) and oceanic magnetic anomalies [*Larson and Pitman*, 1972; *Larson and Hilde*, 1975]. Numeric ages correspond to the time scale of *Van Hinte* [1976]. Arrows and question marks indicate the positions of reversals detected in the Valdorbia (ISEA), Valle del Mis (VM), and Quero-Schievenin (QS) sections that have not yet been satisfactorily confirmed in other sections. This correlation makes anomaly 34 younger than estimated by *Larson and Pitman* [1972] and makes the youngest of the *M* sequence anomalies older than the scheme of *Larson and Hilde* [1975].

val (equivalent to about 1.3 m.y. duration) of 'intermediate directions' in the Cenomanian which they attributed to instability of the geomagnetic field. No rock magnetic or other paleomagnetic evidence support this contention. Corresponding erratic directions are not observed in any of the three sections we have measured which span this interval (Figure 3). The reversals and unstable field behavior which they infer may result from inadequate recording of the paleomagnetic field in this part of the Quero-Schievenin section. The rock magnetic properties of these samples require more extensive investigation before the irregular magnetization can be attributed unequivocally to geomagnetic field behavior. However, our sampling interval in the Cenomanian of the Gubbio and Moria sections was up to 2 m; although closer spaced in the Cismon section, it may be that short Cenomanian reversals were missed.

The absence of additional Albian and Aptian reversals in our sections may relate to the problem of sampling the incoherent Fucoid marls which characterize the lithology in this time interval. Only indurated, more calcareous beds could be drilled, and occasionally, a gap of up to 7 m in the sampling interval was unavoidable. This represents over a million years (Table 1); if reversals were taking place during this time, one (or more) could have been missed in some (or all) of the sections. Therefore we cannot exclude the possibility of other Albian and Aptian geomagnetic reversals. On the other hand, it may be inferred that the indurated beds in the Umbrian sections of the Fucoid marls do not represent stratigraphic time markers.

Reversed Directions in DSDP Cores

Paleomagnetic studies on Deep Sea Drilling Project (DSDP) cores suffer from several serious handicaps. Only the inclinations of magnetization can be used; because of unknown rotations between recovered core segments the declinations are meaningless. Accidental inversion of short recovered segments can falsely imply reversals, and recovery problems lead to distortion of the apparent duration of polarity chrons. Tectonic fold tests cannot be used to indicate age of magnetization.

In spite of these limitations, *Keating et al.* [1975] derived a magnetic stratigraphy for the Late Cretaceous by piecing together results from DSDP sites 207A and 208 and from seven sites in a red horizon of Maastrichtian age in the Mendez Shale. The polarity assignments are not all unambiguous, and no rock magnetic characteristics were given, but the main features of the composite magnetic stratigraphy agree with the anomaly 30–32 anomaly sequence. An interesting aspect was the indication of additional very short reversals for which there is no evidence in oceanic anomaly patterns or in our pelagic limestone sections, in the corresponding Maastrichtian and Campanian intervals. They also cited, but did not show, unpublished evidence of a Cenomanian reversal.

Keating and Helsley [1978] have summarized the paleomagnetic data from legs 40, 41, 43, and 44. Mixed polarities were found in the Maastrichtian and Campanian, recovery was poor in the Santonian to Turonian stages, mixed polarities were reported in the Albian and Aptian and in the older Cretaceous stages. The general nature of the Cretaceous geomagnetic field, with a quiet interval bounded by regions of mixed polarity, is apparent. But the lack of both continuous recovery and precise biozonation make correlation and correct dating of the reversals difficult.

In sediment cores from DSDP site 263, well-defined reversed inclinations were found at three levels [*Green and Brecher*, 1974; *Jarrard*, 1974]. Unfortunately, the dating of these Lower Cretaceous sediments is controversial and depends on the taxa used. Arenaceous and calcareous benthonic foraminifera gave Aptian or older ages; nannoplankton ages correspond to the Upper Albian; dinoflagellate ages are Barremian; and pollen ages are Upper Aptian to Upper Neocomian. In view of these discrepancies, if the site 263 reversed inclinations represent geomagnetic field reversals, they could even correspond to the youngest of the *M* sequence.

Hailwood et al. [1980] have studied the magnetization of Cretaceous sediments drilled during IPOD (International Program of Ocean Drilling) legs 47A, 47B, and 48 in the north Atlantic. The polarity record is discontinuous and most of the samples yield normal polarity. However, four zones of mixed polarity were found. (1) Site 398 (leg 47B) gave some reversed intervals which may be correlated with Campanian and Maastrichtian reversals observed at Gubbio [*Lowrie and Alvarez*, 1977*a*, *b*]. (2) Site 400A (leg 48) gave some very rapid in-

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clination changes in the Late Albian. The same nannofossil zones (*E. turriseiffeli* and *P. cretacea*) were sampled in the Cismon section [*Channell et al.*, 1979] and in the Poggio le Guaine section [*Lowrie et al.*, 1980], but no reversals were found. *VandenBerg* [1979] found two very short reversals (VM, Figure 4) in the equivalent foraminiferal zone (*Tic. breggiensis*) in the Valle del Mis section. It is possible, but as yet not certain, that the reversals at site 400A can be correlated with those found in the Valle del Mis. (3) Site 402A (leg 48) revealed a well-defined inclination pattern interpreted as indicating two zones of reversed polarity close to each other in the late Aptian, *P. angustus* zone. Part of this nannofossil zone was also sampled at Gorgo a Cerbera, Valdorbia, Poggio le Guaine, Cismon, and Cerro Veronese, but no reversals were identified. The uppermost part of the *P. angustus* zone is missing in the Cismon and Cerro Veronese sections, but foraminiferal dating indicates that the corresponding parts of our Umbrian sections are complete. The late Aptian reversals at site 402A are almost the same age as the ISEA reversal (Figure 4) found in Umbria. (4) Site 397 (leg 47A) yields a single Early Barremian reversal which is not inconsistent with the Barremian stratigraphy of the Cismon section.

Summary

The reversals ISEA, VM, and QS in Figure 4 have not yet been well established as a record of geomagnetic field behavior. None of these possible reversals is seen in the pelagic limestone sections in Umbria, or at Cismon and Cerro Veronese in the southern Alps (Figure 3). This may be because our sample spacing in the appropriate intervals in these sections was too wide to locate short reversals and because a small hiatus exists in the Early Albian in our southern Alps sections (Figure 1). DSDP and IPOD cores reveal inclination reversals [*Keating and Helsley*, 1978; *Hailwood et al.*, 1980] which are not always well defined but which may correlate with the ISEA and VM reversals. The dating precision, and in some cases the magnetic data quality, are not sufficient to uniquely confirm this correlation.

A composite magnetic stratigraphy derived from South Tethyan limestone sections (Figure 4) has been made, using only those reversals which we consider to be safely confirmed and omitting the ISEA, VM and QS reversals, whose stratigraphic positions are indicated with arrows and question marks. Future confirmation of these reversals will alter the appearance of the magnetostratigraphic column, but not the identification of the anomalies 29–34 and *M*0–*M*4, which can be directly compared to the polarity history deduced from oceanic magnetic anomalies.

DISCUSSION

In order to compare satisfactorily the polarity sequence recorded in such different media as the basaltic layer of the oceanic crust and a pelagic carbonate formation, it is necessary to refer them to the same time base, an 'absolute' time scale. The only oceanic magnetic anomalies for which numeric ages are unambiguously known are those for which a direct correlation with the magnetic polarity sequence in radiometrically dated Pliocene and Pleistocene lavas is possible. *Heirtzler et al.* [1968] assumed that the sea floor spreading rate in the south Atlantic remained constant, and dated the remainder of the Cenozoic and Late Cretaceous anomalies by extrapolation from this short known correlated interval.

The *M* sequence anomalies are even more poorly dated. The Deep Sea Drilling Project (DSDP) has yielded paleontological ages of the oldest sediments above five Mesozoic anomalies. The ages have rather wide ranges and can only very tentatively be assigned numeric values. The other Mesozoic anomalies were dated by extrapolation and interpolation [*Larson and Pitman*, 1972; *Larson and Hilde*, 1975].

The time scale frequently used as a standard for assigning numeric ages to the principal stage boundaries is that of the *Geological Society of London* [1964]. Each of the Cretaceous stages was assumed to have the same duration of 6 m.y. Although much work has been expended to determine the ages of some of these boundaries radiometrically, they are not yet well know, but the stages apparently do not have equal duration. In a recent compilation, *Van Hinte* [1976] summarizes the Cretaceous biozonation schemes and cities 'best-estimate' numeric ages for the stage boundaries. This time scale has been used in the present paper to allocate numerical values to the Cretaceous stages (Figure 3 and 4).

Geomagnetic polarity is presumed to have only two stable states, and a polarity sequence reduces to a binary string. Fluctuations of sedimentation rate in the pelagic carbonates or in sea floor spreading rate could drastically alter the appearance of the appropriate polarity sequence. Unless an unmistakable pattern of polarities is present—as in the Late Cretaceous—unambiguous correlation of two polarity sequences is not possible. The last of the Mesozoic anomalies, *M*0 to *M*4, form a 'fingerprint.' which is not very long but is sufficiently distinctive to correlate it with a similar 'fingerprint' in the pelagic limestones (Figures 3 and 4).

Older magnetic anomalies than those represented by the polarity sequence in Figure 4 have also been investigated in magnetic stratigraphy studies in carbonate rocks. The paleomagnetism of an excellent section of Lower Cretaceous Biancone limestone has already been measured at Cismon in the southern Alps [*Channell et al.*, 1978]. Although a clear, stable sequence of reversals is preserved in this section through the Hauterivian and Barremian stages, they do not correlate readily with the magnetic anomalies of *Larson and Hilde* [1975] for these stages.

In agreement with the evaluation of Late Cretaceous magnetic and biostratigraphic time scales of *Sclater et al.* [1974] the Cretaceous-Tertiary boundary is located in the reversed interval between anomalies 29 and 30 [*Lowrie and Alvarez*, 1977*a*; *Roggenthen and Napoleone*, 1977], instead of between anomalies 26 and 27 as in the scale of *Heirtzler et al.* [1968]. The Cretaceous magnetic quiet interval ends with anomaly 34 near the Santonian-Campanian boundary, instead of in the Coniacian as estimated by *Larson and Pitman* [1972].

LaBrecque et al. [1977] accepted these correlations, and assuming a constant rate of spreading in the North Pacific in conjunction with their reversal time scale for anomalies 23–29, they determined an age of 80 m.y. for the Santonian-Campanian boundary, slightly older than in the *Van Hinte* [1976] scale used in Figures 3 and 4. Their revision to the *Heirtzler et al.* [1968] time scale resulted in an increase of Cenozoic spreading rates of about 7%.

The youngest anomalies of the *M* sequence are also relocated by these magnetostratigraphic correlations and are older than originally estimated [*Larson and Pitman* 1972; *Larson and Hilde*, 1975]. *M*0 is moved from the Late Aptian to the base of the Aptian, and *M*1 is middle Barremian in age instead of Late Barremian (Figure 4).

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TABLE 3. Average Sea Floor Spreading Rates (cm/yr)

Period	Spreading Center			
	Central Atlantic	South Atlantic	Phoenix-Pacific	Farallon-Pacific
Early Tertiary	1.6	2.1	1.8	4.0
Late Cretaceous	2.6	2.1	4.3	4.7
Cretaceous quiet zone	1.8	3.2	11.3	N4.4 S6.3
Early Cretaceous	1.2	···	5.9	3.2

Recomputed from *Larson and Pitman* [1972] on the basis of the Cretaceous anomaly ages derived from pelagic carbonate rocks (Figure 4).

As a result of these new ages, the duration of the Cretaceous magnetic quiet interval is found to be longer than previously estimated. Spreading rates in this interval are reduced to about 70% of those determined by *Larson and Pitman* [1972]. *Lowrie* [1979] recomputed Cretaceous sea floor spreading rates (Table 3) and found less contrast between rates for the quiet zone and adjacent regions.

However, the rapid increase of spreading rates at all spreading centers in the Atlantic and Pacific areas during the time of the quiet zone, compared to the preceding Early Cretaceous period [*Larson and Pitman*, 1972], appears to be supported by the recomputation. On the other hand, in the central Atlantic and for the Farollon-Pacific plate north of the Mandocino fracture zone the increased rate of spreading persisted into the Late Cretaceous and Early Tertiary, and cannot be considered as a rapid spreading pulse restricted to the quiet zone.

CONCLUSIONS

Magnetic stratigraphy studies in pelagic carbonate rocks have confirmed the main features and many of the details of the history of geomagnetic reversals deduced from oceanic magnetic anomalies between anomaly *M4* in the Early Cretaceous (Barremian) and anomaly 29 at the base of the Paleocene. The anomalies on either side of the Cretaceous magnetic quiet zone have been tied into a biostratigraphic time scale. The quiet zone corresponds to the long interval of constant normal geomagnetic polarity lasting from the time of anomaly *M0* near the Aptian/Barremian boundary to anomaly 34 at the Santonian/Campanian boundary.

Several important features of these magnetic stratigraphy studies require extensive further investigation. The most obvious is to extend the work to younger anomalies in the Tertiary as well as to the older anomalies of the *M* sequence and into the Jurassic magnetic quiet zone. Other features that present interesting topics for further magnetic stratigraphy studies are investigation of the reality of unconfirmed reversals, such as ISEA, VM, and QS (Figure 4) and investigation of the magnetic stratigraphy of the Cretaceous/Tertiary boundary.

The Cretaceous/Tertiary boundary as defined from foraminifera in the Umbrian sequence (extinction of the genus Globotrancana) falls in the negative interval between anomalies 29 and 30 [*Lowrie and Alvarez*, 1977*a*, *b*; *Alvarez and Lowrie*, 1978]. On the other hand, *Butler et al.* [1977] have located it in dinosaur-bearing sections within the positive interval of anomaly 29 or the overlying reversed zone, i.e., distinctively younger than in the pelagic sequences. If both magnetic correlations are correct the discrepancy may represent a difference in the times at which foraminifera and dinosaurs became extinct.

Lerbekmo et al. [1979] recently reported an integrated magnetostratigraphic, biostratigraphic, and geochronological investigation of continental Cretaceous/Tertiary boundary sediments, in which they found the extinction of dinosaurs on land to be synchronous with foraminiferal extinctions in the sea. Unfortunately, the quality of the paleomagnetic data was poor, a secondary overprint was present, and the determination of polarity was made by a dubious, probabilistic criterion. Resolution of this problem is of extreme importance to our understanding of the major events related to the Cretaceous/Tertiary boundary.

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